AAKASH SHARMA

Senior Software Engineer | Python & Django | AI/LLM & Microservices

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SUMMARY

Senior Software Engineer with 8+ years building scalable Python microservices and data platforms with Django and FastAPI across the energy and insurance domains. Cut AWS infrastructure costs by 45%, boosted system scalability by 40%, and reduced downtime by 70%. Currently productionizing agentic AI workflows (LLMs, RAG, LangChain/LangGraph) for market intelligence at Wood Mackenzie.

WORK EXPERIENCE

Senior Software Engineer — **Wood Mackenzie** Gurugram, India | 11/2023 – Present

- Reduced AWS infrastructure costs by 45% by profiling and optimizing Python code and right-sizing EC2, Lambda, and RDS usage.
- Improved backend system scalability by 40% by designing Python microservices with Django and FastAPI using async I/O (asyncio) and Pydantic schema validation.
- Achieved 99% market data processing accuracy by developing an automated data quality validation system in Python/Django, backed by pytest unit and integration test suites.
- Developed an AI-driven system that auto-generates scraper ingestion and processing pipelines using LLM agents, reducing new-scraper onboarding from days to minutes.
- Led the investigation and resolution of critical client-impacting production incidents, coordinating root-cause analysis and fixes to restore service reliability.
- Guide a cross-functional team of 15 engineers through architecture reviews, design discussions, and code quality standards.

Software Engineer — **PasarPolis India Pvt Ltd** Gurugram, India | 11/2021 – 11/2023

- Defined and led the technical architecture strategy for Python/Django web-based solutions, establishing design standards adopted by a 6-engineer team.
- Resolved 100% of major production issues in 50 days against a 90-day mandate by leading team-wide triage and remediation; recognized with the Employee of the Month award.
- Reduced system downtime by 70% and improved overall performance by designing robust, scalable, and secure microservices in Django.
- Delivered ETL pipelines, database solutions, and end-to-end web and API services, orchestrating event-driven data workflows processing 23 million rows per day across 20+ pipelines with Apache Airflow, Celery, and RabbitMQ.

Software Developer — **Knowlarity (Cloud Communications)** Gurugram, India | 06/2021 – 11/2021

- Developed and enhanced agent-status timer functionality for cloud communication workflows.
- Investigated and resolved delayed Call Detail Record (CDR) processing issues and multiple production bugs, improving system reliability.

Software Engineer — **RenewBuy** Gurugram, India | 12/2019 – 06/2021

- Led the design and development of an operations panel in Python/Django as technical lead for an 11-engineer team, cutting manual data entry by 50% and saving 20+ hours per week.
- Designed and developed a video inspection rule engine that achieved 95% fraud detection accuracy.
- Engineered an automated policy renewal journey using Python/Django, contributing to approximately 60% growth in renewal revenue.
- Engineered an automated policy renewal journey in Python/Django that drove substantial growth in policy renewals.

Python Developer — **Bains Innovation Labs** Chandigarh, India | 06/2018 – 12/2019

- Developed Django REST APIs for Autolife and Carsandjobs, enabling seamless integration with third-party platforms.
- Integrated PayPal, Authorize.net, and Moneris payment gateways.

ADDITIONAL EXPERIENCE & TRAINING

Python & Django Development Training Chandigarh, India | 01/2018 – 06/2018

- Completed six months of structured Python and Django training before entering professional software development.

Computer Teacher Jalalabad (West), India | 09/2015 – 12/2017

- Taught computer fundamentals, MS Office, and introductory programming to Classes 10–12, while supporting teachers with digital academic materials.

SKILLS

Languages & Frameworks: Python, Django, Flask, FastAPI, asyncio, Pydantic, RESTful APIs, SQL

AI & LLM: OpenAI, Anthropic Claude, AWS Bedrock, Gemini, RAG, Vector Databases (pgvector, FAISS), Embeddings, Knowledge Graphs, LangChain, LangGraph, MCP (Model Context Protocol), A2A (Agent-to-Agent), AI Agents

Data & Messaging: Apache Airflow, Celery, RabbitMQ, MySQL, PostgreSQL, Redis, MongoDB, DynamoDB, ETL Pipelines

Cloud & DevOps: AWS (EC2, S3, Lambda, SQS/SNS, RDS, CloudWatch), GCP, Docker, Kubernetes, Terraform (IaC), CI/CD, Linux, Git, GitLab

Testing & Observability: pytest, unittest, Integration Testing, CloudWatch, Grafana, Prometheus

Security: OAuth2, JWT

Practices: Microservices, Event-Driven Architecture, System Design, Code Review, Agile/Scrum, Performance Optimization

EDUCATION

FCET Ferozepur — **Bachelor of Engineering and Technology, Computer Science and Engineering**
Ferozepur, India | 07/2011 – 08/2015

AWARDS

- Employee of the Month, PasarPolis (04/2022)